

KIMBALL, NE, USA

WMO: 725665

Lat: 41.189N

Lon: 103.671W

Elev: 1492

StdP: 84.64

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 94073

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-19.4	-16.1	-22.7	0.6	-18.2	-19.8	0.8	-13.6	16.5	3.7	15.1	-0.8	3.9	250	0.666

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	34.6	15.9	32.7	15.9	31.2	15.8	19.1	26.5	18.4	26.1	17.7	25.7	4.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.3	14.9	20.5	16.3	14.0	20.5	15.3	13.1	19.9	61.2	26.1	58.5	26.1	56.0	25.2	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.2	12.0	10.7	DB	-25.2	37.3	3.0	1.4	-27.4	38.3	-29.1	39.1	-30.8	39.9	-33.0	40.9	(4)
(5)			WB	-25.5	20.6	3.0	0.9	-27.6	21.3	-29.4	21.8	-31.0	22.4	-33.2	23.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	-1.6	-1.4	4.2	7.3	12.2	18.6	22.6	20.9	16.4	9.0	3.4	-2.2
	DBStd	9.96	5.99	6.49	5.47	4.77	4.88	3.81	2.92	2.97	4.32	5.27	5.72	6.30
	HDD10.0	1673	358	321	187	104	34	1	0	0	6	81	204	378
	HDD18.3	3652	617	554	437	331	194	43	5	13	84	291	448	636
	CDD10.0	1376	0	1	9	23	103	257	392	338	198	48	6	0
	CDD18.3	314	0	0	0	0	6	49	138	93	27	1	0	0
	CDH23.3	4527	0	0	7	24	169	779	1745	1186	541	74	2	0
	CDH26.7	1941	0	0	0	1	38	317	869	507	199	10	0	0
Wind	WSAvg	4.8	5.3	5.1	5.3	5.5	4.8	4.4	4.0	3.9	4.2	4.8	4.9	5.2
Precipitation	PrecAvg	429	10	8	27	40	72	71	70	45	35	26	14	11
	PrecMax	635	33	23	70	99	200	169	185	121	119	79	38	34
	PrecMin	169	0	0	2	3	15	20	10	1	2	0	1	0
	PrecStd	96	7	6	20	24	41	40	44	28	27	20	9	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.0	19.2	24.2	26.1	30.2	35.9	37.2	35.1	33.5	28.0	22.9	16.1
		MCWB	5.7	7.5	9.7	10.1	13.7	15.3	16.2	15.4	15.5	11.6	9.0	5.6
	2%	DB	12.6	14.9	20.8	23.1	27.4	32.6	35.1	32.9	31.2	25.6	19.8	12.6
		MCWB	4.0	5.3	8.0	9.8	12.9	15.5	16.4	15.8	14.6	11.3	8.2	4.2
	5%	DB	9.8	11.3	17.8	20.6	25.2	30.3	33.2	31.6	29.1	22.7	16.4	9.8
		MCWB	2.9	3.7	6.9	9.0	12.6	15.4	16.3	15.8	14.0	10.3	6.4	3.0
	10%	DB	6.9	7.7	14.9	17.5	22.4	27.8	31.9	29.9	27.0	19.6	12.9	7.1
		MCWB	1.7	2.1	5.8	8.0	11.8	15.1	16.3	16.0	13.4	9.5	4.9	1.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	7.7	10.5	12.6	16.4	19.4	20.2	20.0	18.0	13.8	10.0	6.1
		MCDB	14.4	18.2	20.9	21.8	23.8	26.9	27.0	26.8	26.2	23.0	21.1	14.7
	2%	WB	4.5	5.6	8.8	10.7	15.0	18.2	19.3	18.9	16.7	12.4	8.4	4.7
		MCDB	11.2	13.8	18.8	19.8	22.6	26.2	26.9	25.8	24.4	21.1	18.8	12.0
	5%	WB	3.2	4.0	7.4	9.7	13.8	17.2	18.6	18.0	15.6	11.1	6.8	3.2
		MCDB	8.8	10.5	16.8	19.0	21.7	25.7	26.5	25.2	23.4	20.2	15.6	9.1
	10%	WB	1.9	2.3	6.1	8.5	12.7	16.2	18.0	17.3	14.8	10.0	5.3	1.8
		MCDB	6.6	7.5	14.1	16.7	20.3	24.7	26.2	24.8	23.1	18.5	13.0	6.7
Mean Daily Temperature Range	5% DB	MDBR	12.5	12.8	14.5	14.0	14.2	15.4	16.0	15.6	16.2	15.0	14.3	12.6
		MCDBR	16.0	17.4	19.9	20.5	19.0	19.1	18.9	18.8	19.9	21.2	19.3	16.2
		MCWBR	9.1	9.4	9.8	9.0	7.3	6.5	5.1	5.3	6.8	9.2	9.6	9.3
	5% WB	MCDBR	14.8	16.4	18.6	18.4	15.9	16.5	15.4	15.1	16.0	18.3	18.6	15.2
	MCWBR	9.1	9.3	9.6	8.8	7.0	6.8	5.3	5.5	6.2	8.4	9.4	9.1	
Clear-Sky Solar Irradiance	taub	0.239	0.247	0.266	0.300	0.321	0.325	0.353	0.345	0.307	0.272	0.253	0.240	
	taud	2.608	2.582	2.551	2.450	2.464	2.479	2.433	2.443	2.533	2.602	2.576	2.578	
	Ebn at Noon	955	990	997	973	954	946	915	915	936	945	923	922	
	Edh at Noon	65	78	90	108	109	108	112	108	91	75	66	61	
All-Sky Solar Radiation	RadAvg	2.36	3.20	4.46	5.44	6.25	7.34	7.20	6.33	5.15	3.60	2.56	1.97	
	RadStd	0.10	0.20	0.35	0.38	0.51	0.39	0.28	0.19	0.27	0.29	0.14	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page